Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

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Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No: 06

Experiment Title: Design and implementation of a 2-bit comparator

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Experiment No: 06

Experiment Name:

Design and implementation of 2-bit comparator.

Objective:

- To design and implement a 2-bit comparator.
- To compare two binary numbers
- To indicate the value $A>B$, $A=B$, $A<B$.

Required Components:

Digital trainer board, bread board, connecting wires, IC 7404 (NOT), IC 7408 (AND), IC 7432 (OR), IC 7486 (XOR).

Theory:

A 2-bit comparator compares two bit numbers.

$A = A_1A_0$, $B = B_1B_0$

if $A_1 = B_1$ and $B_0 = A_0$, $x = 1$ and $A = B$

if $A_1 > B_1$ or $A_1 = B_1$ and $A_0 > B_0$, $y = 1$ and $A > B$

if $A_1 < B_1$ or $A_1 = B_1$ and $A_0 < B_0$, $z = 1$ and $A < B$

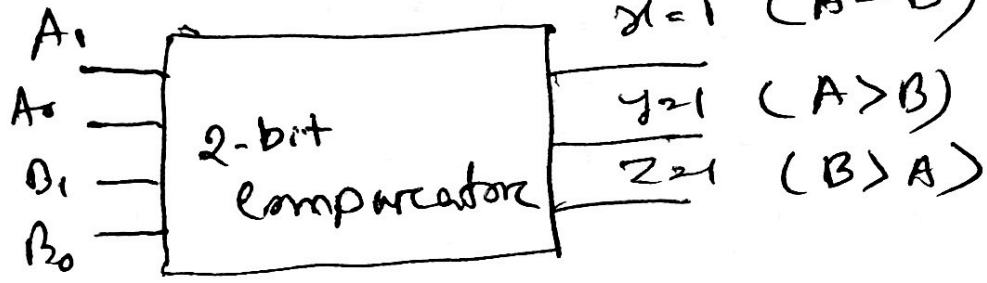
Boolean Expression:

When, $x = 1$ $\quad \overline{A_1 \oplus B_1} \cdot \overline{A_0 \oplus B_0}$

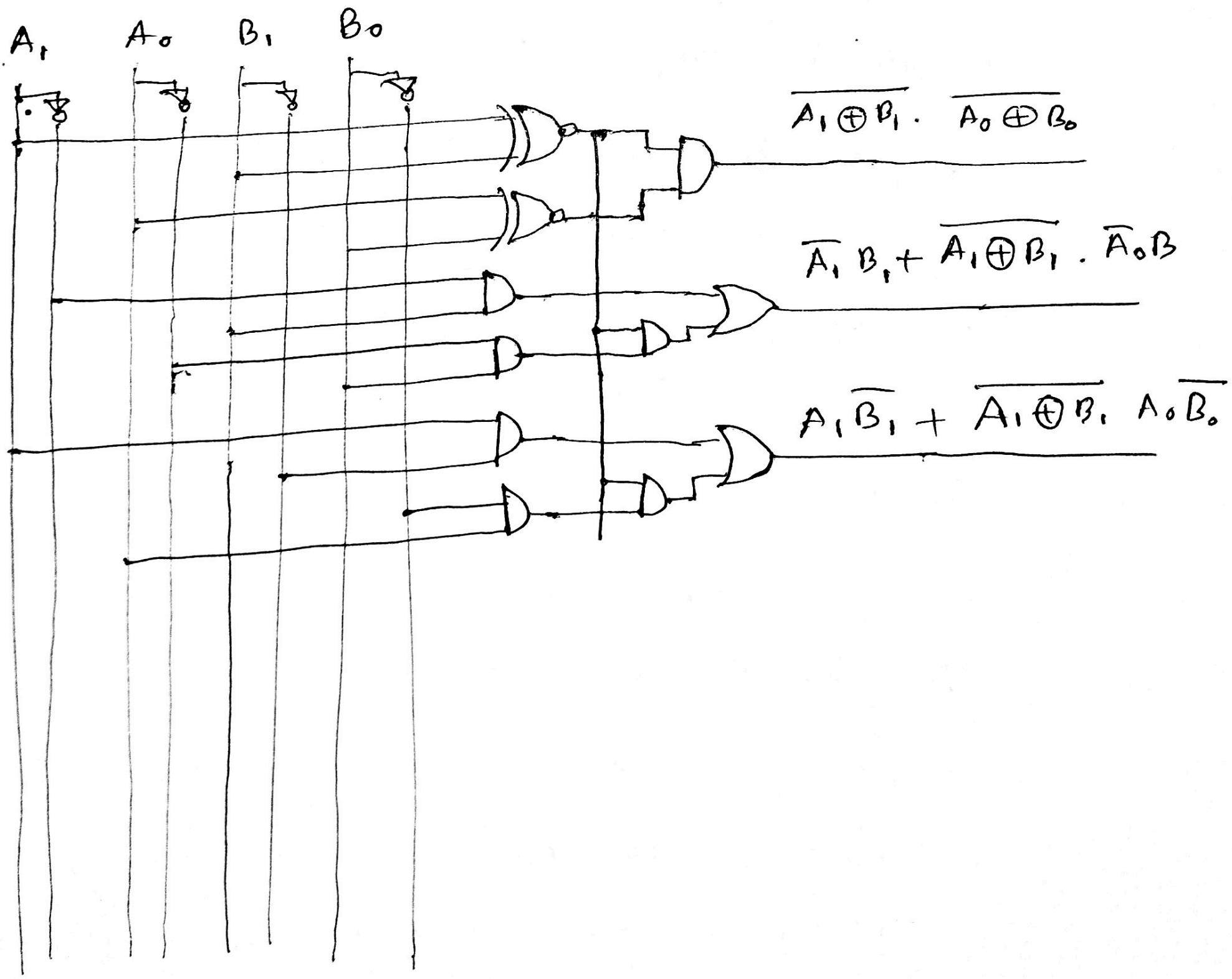
When, $y = 1$ $\quad A_1\overline{B_1} + \overline{A_1 \oplus B_1} \cdot A_0\overline{B_0}$

When, $z = 1$ $\quad \overline{A_1}B_1 + \overline{A_1 \oplus B_1} \cdot \overline{A_0} \cdot B_0$

Block Diagram:



Circuit Diggram:



Procedure:

- Place ICs and Vcc, grounds on the breadboards.
- connect A1, A0, B1, B0 to push-button switches (0/1)
- connect output wires to LEDs.

A_1	A_0	B_1	B_0	x	y	z
0	0	0	0	1	0	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	0	0	1
0	1	1	0	0	0	1
0	1	1	1	0	1	0
1	0	0	0	0	1	0
1	0	0	1	1	0	0
1	0	1	0	1	0	1
1	0	1	1	0	0	1
1	1	0	0	0	1	0
1	1	0	1	0	1	0
1	1	1	0	0	1	0
1	1	1	1	1	0	0

Result / Output:

For, A > B y = 1 LED will be on

For, A = B x = 1 " " "

For, B < A z = 1 " " "

Discussion:

- The circuit correctly compared two bit numbers
- Initially caused false outputs but fixed later.
- Practical results aligned perfectly with theoretical predictions.
- The truth table was matched.